IrriBot – Smart Irrigation Agent

1. Identify the Problem in Context:

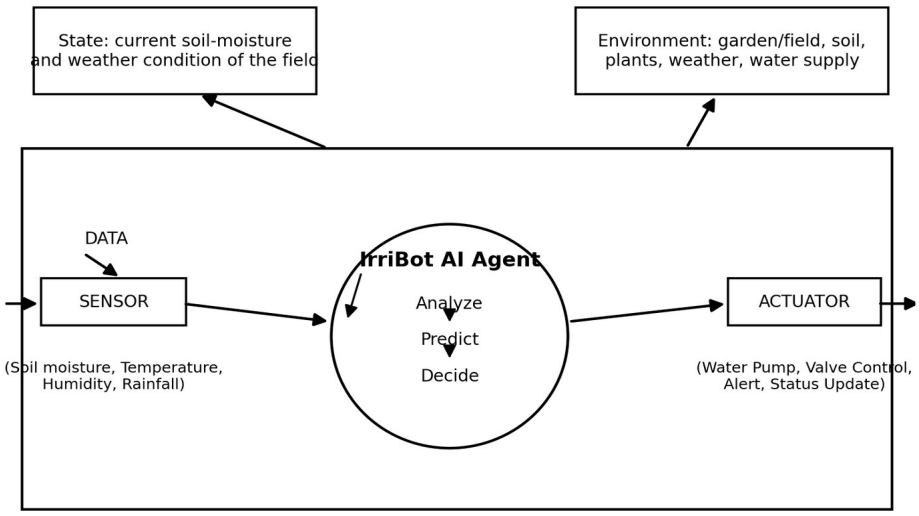
The problem is to design an intelligent AI agent called IrriBot that can automatically manage irrigation for a garden or field based on soil moisture, temperature, humidity and weather conditions. The main problem is that manual watering is often inconsistent because people do not have real-time information about the soil condition and changing weather, which leads to overwatering or underwatering of plants. A fixed watering schedule is not suitable for every day or every season, since water requirements change according to temperature, humidity and rainfall.

2. Problem Formulation:

Textual Representation:

The problem is to design an intelligent AI agent called IrriBot that automatically decides when and how much to irrigate a garden or field based on soil moisture level, temperature, humidity and rainfall. The agent collects this information through its sensors, analyses the current soil and weather condition, predicts whether the plants need water and decides whether to turn the water pump on or off. Its goal is to keep the soil moisture at a healthy level for plant growth while avoiding unnecessary water wastage and maintaining a suitable balance between watering and rest periods. The agent continuously monitors environmental conditions and updates its irrigation decision whenever the soil moisture or weather condition changes.

Graphical Representation



PEAS Framework

P – Performance Measure

IrriBot's performance can be evaluated through:

- Plant health maintained

- Water usage minimised
- Correct soil-moisture level achieved
- Timely irrigation response
- Minimal manual intervention needed
- User satisfaction with the watering routine

E – Environment

- Garden or agricultural field
- Soil condition
- Plant type and water requirement
- Weather conditions
- Water supply system
- Time of day / season

A – Actuators

- Water pump ON/OFF
- Control valve open/close
- Status display
- Alert notification
- Irrigation schedule adjustment
- Plan modification

S – Sensors

- Soil moisture sensor
- Temperature sensor
- Humidity sensor
- Rain sensor
- Water flow sensor

Environment Types

IrriBot can be evaluated using different AI environment types:

Environment Type	IrriBot
Partially Observable	The agent cannot directly observe deeper root-zone moisture or the plant's complete internal condition — only surface-level sensor readings.
Stochastic	Rainfall, temperature and evaporation rates change unpredictably, so the same watering action can lead to different outcomes.
Sequential	Today's irrigation decision affects soil moisture and plant health in the following days, so past actions influence future states.
Dynamic	Weather and soil conditions keep changing even while the agent is deciding, independent of the agent's own actions.

Environment Type	IrriBot
Continuous	Soil moisture, temperature and humidity are continuous numeric values rather than a fixed finite set of states.
Single Agent	IrriBot works alone to manage irrigation for the field; there is no other agent competing or cooperating for the same task.